

# Kimmeng HONG

+33 744771289 | [hongkimmeng17@gmail.com](mailto:hongkimmeng17@gmail.com) | [in Kimmeng HONG](#) | [Kimmeng007](#) | [Portfolio](#) |  
Ivry-sur-Seine, France

## PROFILE

Master's graduate in Data Science (Université Paris-Saclay, defense pending October 2026), with experience across classical machine learning, deep learning and computer vision (YOLO, biometric verification), and LLM-based systems (RAG, agentic AI). Comfortable building and deploying models, from data preprocessing to production-facing APIs.

## EXPERIENCE

- **Audensiel** 

Apr 2026 – present  
AI Engineer Intern  
Boulogne-Billancourt, France

  - Built a synthetic multi-modal flood-rescue dataset (RGB, depth, segmentation) in Unreal Engine 5.5 / Cosys-AirSim across varying water levels, weather, and altitude.
  - Improved victim detection (YOLO fine-tuning across Human/Vehicle/Boat classes, +12.3% Human-class AP50), trained a semantic segmentation model (PSPNet, MobileNetV2, ~0.94 mean IoU) for water/ground/road/building classification, and built a real-time perception pipeline (multi-object tracking, LiDAR-camera 3D fusion, interpretable risk scoring).
  - Developed a live web dashboard (FastAPI) streaming annotated video during flights.
- **AI Farms Co., Ltd.** 

May 2025 - Aug 2025  
AI Engineer Intern  
Phnom Penh, Cambodia

  - Developed a proof-of-concept vision-based speaker detection system using video input, modeling spatial and temporal patterns in face sequences with a CNN-LSTM architecture.
  - Built a reproducible pipeline (Python, PyTorch, OpenCV, YOLOv8) for preprocessing, feature extraction, training, and evaluation.
  - Evaluated CNN backbones (ResNet-50, EfficientNet-B0) on 3,786 labeled video scenes, achieving up to 72.95% F1-score on a held-out test set.

## EDUCATION

- **Université Paris-Saclay**

Sep 2025 - Expected Nov 2026  
Master's Degree (M2) in Data Science: Health, Insurance and Finance  
Évry, France
- **École Nationale Supérieure d'Informatique pour l'Industrie et l'Entreprise (ENSIIE)**

Sep 2024 - Expected Nov 2026  
Double Engineering Degree in Applied Mathematics (ITC-ENSIIE)  
Évry, France
- **Institut de Technologie du Cambodge (ITC)**

Oct 2020 - Jul 2025  
Engineering Degree in Data Science  
Phnom Penh, Cambodia

## SELECTED PROJECTS

- **Multimodal Biometric Verification for Mobile Banking KYC**

Python, PyTorch, InsightFace, SpeechBrain 

  - Built a face + voice 1:1 identity verification system for mobile-banking KYC using pretrained ArcFace and ECAPA-TDNN embeddings, achieving 0.27% EER on face and 2.26% EER on speaker verification, with score-level fusion roughly halving error rates under simulated low-quality mobile capture.
- **FinDocQA: Agentic RAG for SEC Financial Filings**

Python, LangGraph, Gemini, MongoDB, Streamlit 

  - Built and deployed a retrieval-augmented QA system over SEC 10-K/10-Q filings (LangGraph agent, hybrid dense + BM25 + reranker retrieval), evaluated against the public FinanceBench benchmark; diagnosed and fixed real retrieval failures via a RAGAS evaluation harness, improving context recall by +67%.
- **Sorbonne Data Challenge: Malware Detection**

Machine Learning, XGBoost, Feature Engineering, SVD 

  - Ranked 7th out of 85 teams in a malware detection challenge (Sorbonne University), building an XGBoost classifier on control-flow-graph data across 453 malware families (macro F1 = 0.59).

## SKILLS

- **Programming Languages:** Python, R, C++, Java, SQL, Git
- **Libraries:** PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy
- **Machine Learning:** Deep learning, computer vision, transformers, NLP, regression, classification, clustering, time series forecasting, survival analysis
- **Data & Visualization:** Data cleaning, feature engineering, pipeline design, Matplotlib, Seaborn, Power BI
- **LLMs & Deployment:** LangChain, LangGraph, RAG, LLM API, MongoDB, Docker, CI/CD, FastAPI, Streamlit, MLflow
- **Languages:** English (C1), French (B1), Khmer (native)

## HONORS

**Sophie Germain Scholarship** (FMJH, 2024) , merit-based, for a Master's in France; also awarded the Eiffel Excellence Scholarship (Campus France), declined in favor of Sophie Germain.